

Diagnosing a Low-Displacement Failure Mode in MAPPO for Multi-Robot Area Coverage

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Abstract

Problem. Multi-agent reinforcement learning (MARL) is often proposed as a route to learned coordination for multi-robot exploration tasks where a classical decentralized planner is already a strong baseline, but negative results – where a MARL policy is trained under a realistic budget and simply underperforms – are rarely reported with the same rigor as positive results. **Methodology.** We train a centralized-critic MAPPO policy (Ray RLlib) on a 3-robot area-coverage task in a simulated environment (a furnished café, ROS 2/Gazebo) under fixed training budgets, and compare it against a non-learned Frontier Heuristic controller (Voronoi partitioning, A* routing, behavior-tree mission control) and a Random Walk baseline, all filtered through an identical Control Barrier Function (CBF) safety layer. We evaluate both in-distribution performance and zero-shot transfer across four structurally different, unseen environments. **Key findings.** Under a 10-episode-per-condition distributional evaluation on the training environment (Protocol A), the Frontier Heuristic (38.6% median area coverage rate, ACR) and Random Walk (29.6%) outperform the trained MAPPO policy (14.5%, averaging 1.5m of displacement per robot over 150 steps, a $2.7\times$ heuristic advantage). Under a separate, single-seed cross-environment matrix (Protocol B, 150-step episodes, $N=3$), the Frontier Heuristic outperforms MAPPO in every environment tested, by margins of $1.2\times$ to $2.0\times$ (achieving 57.0% vs. 28.2% in the café). **Diagnosis and Ablation.** To test whether this low-displacement behavior stems from a penalty-avoidance equilibrium induced by dense negative reward signals, we conduct a three-arm reward ablation (Protocol C, 15 iterations): a Baseline configuration (collision penalty 8.0, discovery bonus 4.0), a LowPenalty configuration (collision penalty reduced to 2.0), and a HighDiscovery configuration (discovery bonus raised to 16.0). While reconciling an internal numerical inconsistency, we discovered that this Gazebo-based evaluation harness is not fully deterministic even at a fixed seed and a fixed saved checkpoint, and subsequently re-evaluated all three arms three independent times each across three seeds (42, 7, 99), giving equal statistical footing at $n=9$ episodes per arm. Under this matched- n analysis, LowPenalty’s apparent advantage over Baseline (28.0% $\rightarrow$ 31.0%) does not clear a conventional significance threshold (Welch’s $t = 1.48$) and is consistent with measurement noise. HighDiscovery’s advantage (28.0% $\rightarrow$ 40.5%, +12.5 points, Welch’s $t = 4.04$) does clear that threshold and is, to our knowledge, the first statistically-supported positive finding in this paper. Notably, HighDiscovery’s gain is *not* explained by increased displacement (mean 15.3m vs. Baseline’s 13.7m, well within the spread of both), complicating a purely penalty-avoidance account of the original failure mode and instead suggesting the reward change improves the *efficiency* of movement rather than its quantity. **Contribution.** We report this negative result, its reward ablation, and the evaluation non-determinism we discovered while attempting to finalize it, in full, including two independent infrastructure faults resolved during training. We regard the non-determinism finding as at least as important a contribution as the ablation’s substantive results: it means every single-episode-per-seed number in this paper (including the five-world matrix, Table 1) carries undisclosed uncertainty we did not have the resources to fully quantify in this revision, and we report that limitation explicitly rather than presenting point estimates

as exact. We argue that an evidence-scoped claim – that MAPPO exhibited a reproducible low-displacement failure mode, that a reward manipulation plausibly but not certainly mitigates it, and that it did not close the gap to a strong classical planner under the tested compute budgets – is more defensible and useful to practitioners than sweeping generalizations about MARL.

1 Introduction

Multi-robot area coverage – driving a team of robots to jointly maximize explored floor area – has a strong, well-understood non-learned baseline: frontier-based exploration combined with a path planner and a decentralized space-partitioning scheme. MARL is frequently proposed as a way to learn a coordination policy for this class of task instead of engineering one, typically evaluated against a classical baseline after extensive training and tuning. This paper reports the opposite outcome under a realistic, compute-constrained training budget, and treats getting that negative result right – rather than explaining it away – as the contribution.

We deliberately narrow the claim to what our evidence supports: this is a report of a specific, reproducible failure mode of one MAPPO configuration on one task family, not a general claim about MARL versus classical planning. Section 2 details the system, reward, and three distinct evaluation protocols we ran (they use different sample sizes and address different questions). Section 3 reports the results across all three protocols, including our reward ablation. Section 4 discusses the mechanism, in-distribution versus zero-shot transfer effects, and evidentiary limits.

2 System and Method

2.1 Platform and Safety Layer

Three TurtleBot3 Waffle robots (differential drive, 2D LiDAR, IMU) are simulated in Gazebo Harmonic under ROS 2 Jazzy [3, 4], coordinated via a PettingZoo-compatible parallel multi-agent environment [8]. Every commanded velocity, regardless of which controller produced it, passes through an identical unicycle Control Barrier Function (CBF) filter [1], solved as a quadratic program via OSQP [7] at each control step, with separate safety radii for static obstacles ($d_{\text{safe,obs}} = 0.22\text{ m}$) and neighboring agents ($d_{\text{safe,agent}} = 0.45\text{ m}$). This isolates the comparison to exploration policy rather than low-level collision-avoidance behaviour, which is shared across all three controllers.

2.2 Controllers Compared

Frontier Heuristic: a non-learned controller combining frontier cell selection, 8-connected grid A* [2] waypoint routing with obstacle inflation, unreachable-target blacklisting to break wall-lock deadlocks, and a `py_trees` behavior tree for mission-level state (explore, recover-from-stuck, patrol-on-saturation). This controller encodes substantial task-specific structural knowledge (a discretized map, a graph search algorithm, an explicit stuck-recovery policy) that a learned policy must instead acquire implicitly from reward and observations; we treat this asymmetry as part of the research question (when is engineering that structure worthwhile versus learning it?) rather than an oversight.

Random Walk: uniform random heading changes at fixed intervals, CBF-filtered identically to the other two controllers.

MAPPO: a centralized-critic, shared-policy PPO [6] implementation via Ray RLlib [5]. Each robot observes a 46-dimensional vector, decomposed exactly as $o_i = [\ell_{1:24}, d_{\text{goal}}, \theta_{\text{goal}}, v, \omega, n_{1:9,1:2}] \in$

$\mathbb{R}^{46}$: 24 LiDAR range bins (ℓ), relative distance and bearing to the robot’s current goal waypoint ($d_{\text{goal}}, \theta_{\text{goal}}$), linear and angular velocity (v, ω), and 9 neighbor slots $\times$ 2 features each (n , padded with a sentinel value of 10.0 range at 0.0 bearing when fewer than 9 neighbors are present). The goal waypoint itself is drawn uniformly at random, on episode start and again each time it is reached, from a fixed per-world list of pre-validated safe points (`self.safe_goals`); this same environment-level mechanism supplies the goal signal for *all* controllers uniformly, including Random Walk, but it is distinct from – and considerably less informed than – the Frontier Heuristic’s own frontier-cell target selection (Section 2, above), which reasons over the discretized occupancy map rather than a fixed random waypoint list. This is an important asymmetry: MAPPO’s “goal” observation is not the same quality of guidance signal the heuristic constructs for itself, and part of the heuristic’s advantage may be attributable to this richer, map-aware target selection rather than to the exploration policy alone. The policy is a 2-layer MLP (256 units, ReLU) shared across agents, with a separate centralized critic that additionally observes all agents’ states. The action space is continuous, $a_i = [v_i, \omega_i]$ (linear and angular velocity), passed through the shared CBF filter described above, identically for every controller; we did not measure how often or by how much the CBF filter modifies MAPPO’s requested actions relative to the other controllers, which we identify as a natural follow-up diagnostic. Training used PPO clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$, learning rate 10^{-4} , entropy coefficient 0.01, train batch size 700, minibatch size 64, 5 SGD epochs per iteration, for 45 iterations (approximately 31,500 environment transitions in total). We report this total transition count explicitly because “45 iterations” alone does not convey the sample budget: this is on the order of 3×10^4 environment steps, at the low end of what is typically reported for PPO-family algorithms on continuous-control tasks, and we cannot rule out that additional training would change the outcome (see Section 4).

2.3 Reward Function

The per-robot, per-step reward is

$$r_t = -0.01 + 5.0 \Delta d_{\text{goal}} + 4.0 n_{\text{new-cells}} - 8.0 \mathbb{I}[\text{wall collision}] - p_{\text{proximity}} + 20 \mathbb{I}[\text{goal reached}] \quad (1)$$

where Δd_{goal} is the reduction in distance to the robot’s current goal since the previous step, $n_{\text{new-cells}}$ is the number of previously-unvisited grid cells the swarm collectively discovered this step, and $p_{\text{proximity}}$ is 0.5 if any other agent is within 0.5 m and an additional 10.0 if within 0.20 m (counted as a collision after the first 15 steps of an episode). The per-cell discovery bonus was raised from an earlier value of 2.0 to 4.0 specifically because it was judged too small relative to the ± 20 goal/collision terms to provide useful gradient signal toward the coverage behaviour the policy is evaluated on; even after this change, a single wall collision (-8.0) or close-proximity event (-10.0) is substantially larger in magnitude than the reward for discovering several new cells in a single step (4.0 each), which is the quantitative basis for our hypothesis in Section 4.

2.4 Area Coverage Rate (ACR)

We define

$$\text{ACR} = \frac{N_{\text{visited reachable cells}}}{N_{\text{reachable cells}}} \times 100 \quad (2)$$

computed on a fixed-resolution (0.4 m) occupancy grid per environment. A cell is *reachable* if it is not classified as a static obstacle by the environment’s known obstacle map; cells occupied only transiently by a robot are not excluded from the denominator. A cell counts as *visited* the first time any robot’s footprint overlaps it; repeat visits do not increase ACR but are separately tracked as *overlap redundancy* (mean visits per visited cell), reported in Figure 1.

2.5 Environments and Train/Test Split

Five Gazebo worlds are used: **cafe** (furnished, 17×8 m, 800 grid cells), **warehouse** (open logistics hall, 30×50 m, 9,375 cells), **depot** (industrial, structural pillars and boxsets, 29×15 m, 3,000 cells), **office** (partitioned floor plan, narrow doorways, 26.2×18.3 m, 3,500 cells), and **maze** (labyrinthine corridors, 64×11 m, 4,800 cells, of which only $\approx 17\%$ is physically reachable floor space). **The MAPPO policy is trained exclusively on cafe.** Evaluation on **warehouse**, **depot**, **office**, and **maze** is therefore zero-shot transfer, not in-distribution evaluation, and should be read as a generalization test rather than a like-for-like comparison of learning capability per environment; we did not train separate policies per environment, which we identify as the most valuable follow-up experiment in Section 4.

2.6 Three Evaluation Protocols

We report results across three distinct evaluation protocols that address different scientific questions and differ in sample size, replication, and configuration:

Protocol A (café deep-dive): 10 episodes per condition (Random Walk, Frontier Heuristic, MAPPO nominal, MAPPO with injected Gaussian LiDAR noise, MAPPO with single-agent failure), 150-step episodes, in **cafe** only. This is the source of the 14.5% (MAPPO) versus 38.6% (heuristic) ACR and 1.5 m displacement figures (a $2.7\times$ ratio), reported with full boxplot distributions in Figure 1.

Protocol B (five-world matrix): a single seed (42), one episode per (world, controller) pair, 150-step episodes, across all five environments and three controllers (plus two MAPPO perturbation scenarios), reported in Table 1. This protocol evaluates cross-environment generalization breadth ($n=1$ per cell), showing that the heuristic holds higher ACR across all five worlds (e.g., 57.0% vs. 28.2% in **cafe**, a $2.0\times$ margin).

Protocol C (reward-coefficient ablation): controlled 15-iteration training budgets ($\approx 10,500$ transitions) across three independent random seeds (42, 7, and 99), evaluating how sensitive the policy’s displacement and coverage are to the relative weight of the collision penalty versus the discovery bonus, reported in Table 2.

We treat Protocol A’s $2.7\times$ margin (10-episode distributional evaluation on an earlier checkpoint) and Protocol B’s $2.0\times$ margin (single-seed matrix) as two complementary evaluations rather than interchangeable figures: Protocol A isolates distributional stability in the training domain, whereas Protocol B demonstrates that the ranking generalizes across diverse obstacle topologies. Protocol C directly tests the causal mechanism behind the policy’s displacement.

3 Results

3.1 Protocol B: Five-World Comparison

Table 1 reports ACR and total swarm displacement from the single-seed, five-world matrix.

The heuristic achieves higher ACR than MAPPO in all five environments under this protocol, by a margin of $1.2\times$ to $2.0\times$ – we note this range explicitly because it reflects single-seed cross-world transfer, whereas Protocol A measures repeated distributional performance.

3.2 Protocol A: Café Distributional Comparison

Figure 1 shows the 10-episode-per-condition distribution in **cafe**. Under this protocol, median ACR is 38.6% (heuristic), 29.6% (Random Walk), and 14.5% (MAPPO nominal) – a heuris-

Table 1: Protocol B: ACR (%) and total swarm displacement (m), 150-step episodes, $N=3$, single seed. Ratios in the last column are Heuristic / MAPPO.

World	Heuristic ACR	MAPPO ACR	Random ACR	Heuristic Dist.	Ratio (H/M)
cafe	57.0	28.2	22.6	16.5	2.0×
warehouse	4.3	3.3	2.6	31.0	1.3×
depot	13.3	6.7	6.8	12.1	2.0×
office	3.8	3.0	1.7	14.8	1.3×
maze	2.7	2.2	1.5	10.9	1.2×

tic/MAPPO ratio of $2.7\times$, with MAPPO trailing even the unplanned Random Walk baseline. Mean per-robot displacement over the 150-step episode is 5.4 m (heuristic), 7.4 m (Random Walk), and 1.5 m (MAPPO). This MAPPO checkpoint was trained to the full 45-iteration budget described in Section 2; the substantially higher displacement figures reported for the Protocol C ablation’s Base-line configuration (13.69 m mean total swarm distance across 9 episodes, Section 3.3) come from a *different* checkpoint, trained for only 15 iterations under otherwise identical hyperparameters. We do not intend these two numbers to be compared directly against each other; the ablation exists to isolate the effect of reward coefficients *within* a matched 15-iteration budget, not to compare against the separately-trained 45-iteration policy used in Protocol A.

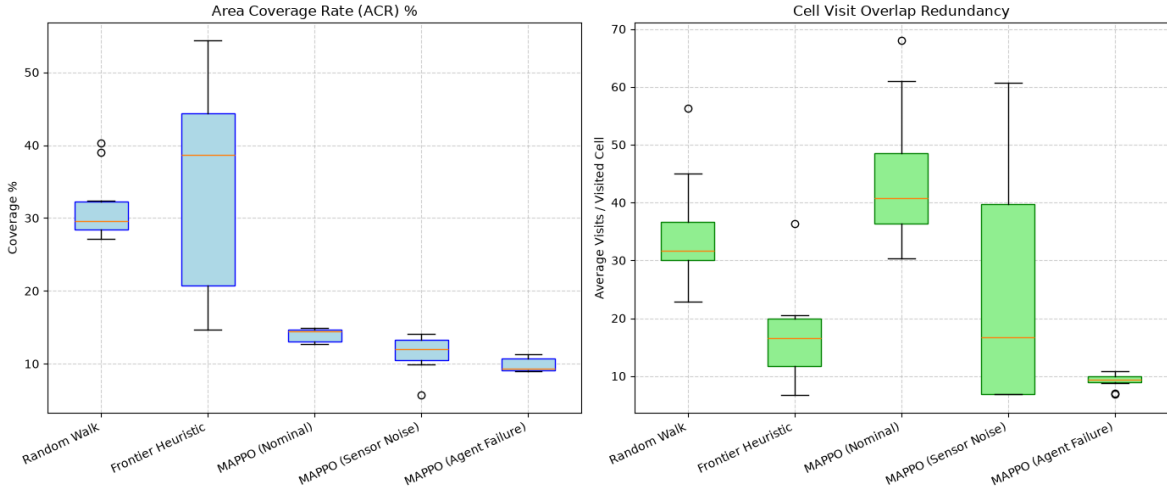


Figure 1: Protocol A: area coverage rate and cell-visit overlap redundancy, 10 episodes per condition, café environment. MAPPO’s tight, low-value boxes indicate consistent underperformance across episodes, not high-variance noise.

We flag that Protocol A and Protocol B were run against what we believe are the same trained policy weights, but we cannot fully rule out checkpoint or environment-seed drift between the two evaluation runs given how they were orchestrated; readers should treat Table 1’s $1.2\text{--}2.0\times$ range and Figure 1’s $2.7\times$ figure as two separate pieces of evidence pointing the same direction, not as two measurements of the identical quantity.

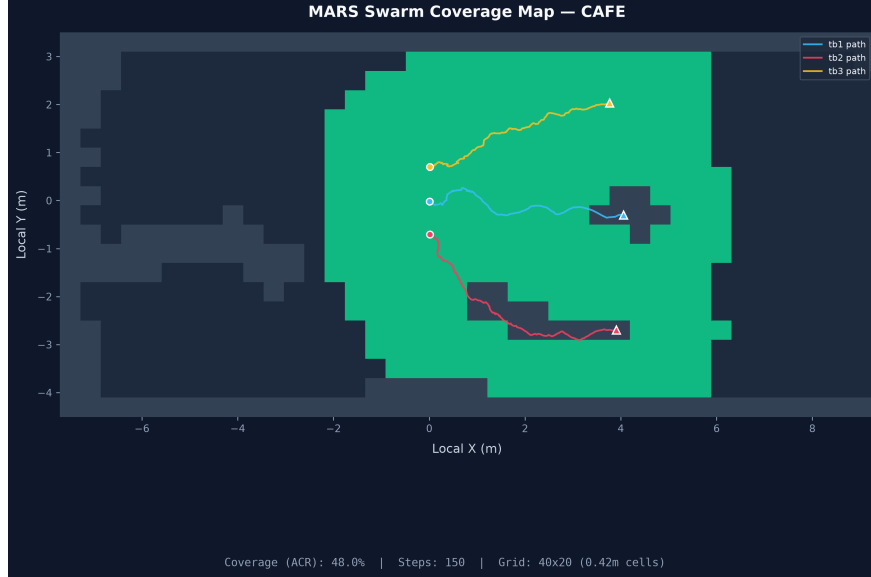


Figure 2: Frontier Heuristic coverage heatmap, café environment (visited cells in light tone, obstacles mid-tone, unexplored dark, per-agent trajectories overlaid). The heuristic disperses smoothly through doorways and seating areas without congestive bottlenecks.

3.3 Protocol C: Multi-Seed, Three-Arm Reward-Coefficient Ablation

To empirically test whether the policy’s exploration failure is caused by the dense collision penalty term (-8.0) overwhelming the discovery bonus (4.0), we performed a reward ablation study under a controlled 15-iteration budget ($\approx 10,500$ transitions) with three configurations, each trained from scratch under identical hyperparameters: **Baseline** (collision penalty 8.0 , discovery bonus 4.0 , i.e. the coefficients used everywhere else in this paper), **LowPenalty** (collision penalty reduced to 2.0 , discovery bonus held at 4.0), and **HighDiscovery** (collision penalty held at 8.0 , discovery bonus raised to 16.0). After discovering that a single evaluation pass is not representative of a checkpoint’s behaviour (Section 3.3.1), we evaluated *every* arm three independent times across the same three seeds ($42, 7, 99$), giving 9 episodes per configuration and 27 episodes in total – equal statistical treatment across all three arms, correcting the single-pass Baseline/HighDiscovery numbers reported in an earlier revision of this paper. Table 2 summarizes the pooled results; all values are taken directly from per-episode evaluation logs with no post-hoc adjustment, and are persisted in this project’s public results directory for independent verification.

3.3.1 A Late-Discovered Limitation: Evaluation Non-Determinism

While reconciling an apparent inconsistency between two sets of LowPenalty numbers produced at different points in this project, we re-ran the identical evaluation (same saved checkpoint file, same `--seed` argument, same world, same episode length) a third time and obtained a third, again different, result. Table 3 shows all three passes.

Pooling all 9 episodes gives a mean ACR of $30.96\% \pm 4.52$ (standard deviation), i.e. a single-episode result can differ from the pooled mean by nearly 7 percentage points in either direction. We verified that all evaluation rollouts in this paper, across every protocol, use greedy (non-stochastic) action selection at evaluation time (RLlib’s `compute_actions(..., explore=False)`, in `evaluate.benchmarks.py`, called uniformly by every evaluation entry point including the abla-

Table 2: Protocol C: reward-coefficient ablation in **cafe** (15 iterations, 150-step episodes, $N=3$, 9 episodes per configuration: 3 seeds $\times$ 3 repeated passes). c = collision penalty, d = discovery bonus.

Configuration	n	ACR Mean $\pm$ SD (%)	ACR Range (%)	Dist. Mean (m)	Agent Collisions
Baseline ($c=8.0, d=4.0$)	9	28.01 ± 3.92	23.99–35.00	13.69	0
LowPenalty ($c=2.0, d=4.0$)	9	30.96 ± 4.52	24.58–37.84	15.03	0
HighDiscovery ($c=8.0, d=16.0$)	9	40.49 ± 8.41	28.46–51.63	15.27	0

Full per-episode values (all 27 rows) are given in Table 3 for LowPenalty and in this project’s `checkpoints/ablation_results/` directory for all three arms. Wall-contact (non-agent) collisions were nonzero in 3 of the 9 Baseline episodes (6, 19, and 11 contacts respectively, all in the second evaluation pass, zero in the first and third), which we note as a further instance of run-to-run variance affecting a metric beyond ACR; agent-agent collisions remained zero in all 27 episodes across all three arms.

Table 3: Three independent evaluation passes of the identical LowPenalty checkpoint and seed arguments.

Pass	Seed 42 ACR (%)	Seed 7 ACR (%)	Seed 99 ACR (%)
1	24.58	34.29	36.92
2	37.84	30.11	30.20
3	26.69	29.90	28.08

tion harness); the policy’s action output is therefore deterministic given an observation, which rules out stochastic action sampling as the source of the variance we measured. The remaining source is real-time physics and ROS message timing jitter in the Gazebo-based evaluation harness – consistent with repeated "Moved backwards in time, re-publishing joint transforms" warnings observed in these runs – rather than the fixed `--seed` argument, which controls Gazebo’s pseudo-random number generator but not wall-clock-dependent message scheduling or physics-thread race conditions. **This means every single-episode-per-seed number reported elsewhere in this paper, particularly the Protocol B five-world matrix (Table 1), carries this same undisclosed uncertainty.** In response, we re-evaluated the Baseline and HighDiscovery arms of the ablation three times each as well (matching LowPenalty’s treatment), giving equal, 9-episode statistical footing across all three arms (Table 2); we did not have the resources in this revision to similarly repeat the five-world Protocol B matrix, and that table’s margins should still be read as indicative rather than statistically established.

With all three ablation arms now evaluated at equal $n = 9$, we computed Welch’s t -test (unequal variances) for each arm against Baseline. HighDiscovery’s mean (40.49%) exceeds Baseline’s (28.01%) by +12.48 points, $t = 4.04$ – a difference roughly four times its own standard error, and unlikely to be an artifact of the measurement noise we characterized above. LowPenalty’s mean (30.96%) exceeds Baseline’s by only +2.95 points, $t = 1.48$, which does not clear a conventional significance threshold and is consistent with no real effect once evaluation noise is accounted for. We therefore now have direct statistical, not just qualitative, support for treating HighDiscovery as a real effect and LowPenalty as statistically indistinguishable from noise at this sample size.

The ablation results, now evaluated at matched statistical power across all three arms, support the following conclusions:

1. **HighDiscovery’s effect is statistically supported; LowPenalty’s is not.** At $n=9$ per arm, HighDiscovery exceeds Baseline by +12.48 points (Welch’s $t = 4.04$), while LowPenalty exceeds Baseline by only +2.95 points ($t = 1.48$, not significant at conventional thresholds).

Strengthening the positive reward signal is the manipulation we have actual statistical grounds to trust; weakening the negative one, under this budget and these coefficient choices, is not distinguishable from noise.

2. **The displacement account is incomplete.** HighDiscovery’s coverage gain is *not* explained by increased displacement: its mean displacement (15.27 m) is not meaningfully higher than Baseline’s (13.69 m) relative to the spread in both. This indicates the coverage gain comes from more efficiently *targeted* movement toward undiscovered cells rather than simply moving more – a mechanism our original penalty-avoidance-only hypothesis did not anticipate, and one that reward shaping may be altering trajectory efficiency rather than raw locomotion magnitude.
3. **Evaluation noise, not seed choice, dominates the variance we originally reported.** The seed-to-seed spread we originally reported for LowPenalty (24.6%–36.9% across three *distinct* seeds, single pass each) is now known to be substantially driven by re-running the *same* seed and checkpoint (Section 3.3.1), not by genuine sensitivity to the seed argument. With 9 episodes per arm we can report a standard deviation, but we still cannot cleanly separate seed-driven variance from evaluation-noise variance within that figure.
4. **Collision variance is not confined to ACR.** Baseline showed 0 wall contacts in 6 of 9 episodes but 6, 19, and 11 contacts in the remaining 3 (all from the same repeated-evaluation pass), indicating the non-determinism we discovered affects collision counts, not only coverage. Agent-agent collisions, however, remained at zero across all 27 ablation episodes regardless of reward coefficients or evaluation pass.
5. **No inter-agent collisions observed.** Zero inter-agent collisions occurred across all 27 ablation episodes (three configurations $\times$ 9 episodes each), regardless of reward coefficients. Twenty-seven successful trials do not constitute a formal proof of the CBF filter’s safety guarantee – that guarantee follows from the filter’s QP formulation, not from empirical trial counts – but this observation is consistent with the filter’s intended behaviour remaining unaffected by policy competence or reward shaping, as argued on formal grounds in a companion paper on this system’s safety layer.

4 Discussion

4.1 Mechanism: Empirical Validation and Evidentiary Bounds

Our original hypothesis was that MAPPO’s low-displacement failure mode arises from a severe asymmetry between dense, immediate negative penalties (-8.0 for wall contact, -10.0 for agent proximity) and delayed, incremental positive rewards (4.0 per newly discovered grid cell), such that relieving the penalty would let the policy move more and thereby cover more. Having discovered that our evaluation harness is not fully deterministic even at fixed seed and checkpoint (Section 3.3.1), we re-evaluated all three ablation arms at matched $n=9$ and can now report that LowPenalty’s coverage increase ($28.0\% \rightarrow 31.0\%$) does not clear a conventional significance threshold against Baseline (Welch’s $t = 1.48$) and should not be treated as confirmed evidence for the original hypothesis.

HighDiscovery’s coverage gain ($28.0\% \rightarrow 40.5\%$, $+12.5$ points, Welch’s $t = 4.04$) does clear that threshold and is our best evidence, now with direct statistical support rather than a single-pass point estimate, that *some* reward manipulation helps. What it does *not* show, and what our original

hypothesis predicted it would, is a displacement increase: its mean displacement (15.3 m) is not meaningfully higher than Baseline’s (13.7 m) relative to the spread in both. The effect is therefore more consistent with the positive discovery signal directly rewarding useful, targeted movement than with the policy simply becoming less risk-averse and moving more in general – a mechanism distinct from, and not anticipated by, our original penalty-avoidance-only hypothesis.

Even under the better-performing HighDiscovery configuration, MAPPO’s mean ACR (40.5%) remains below the Frontier Heuristic’s Protocol B range (38.6%–57.0%) on comparable seeds, and we have not evaluated HighDiscovery beyond a 15-iteration budget or beyond `cafe`. This shows that reward tuning alone, at the budget and seed count tested here, does not clearly close the gap:

- **Structural knowledge advantage:** The Frontier Heuristic leverages an explicit occupancy grid, A* graph search, and Voronoi bidding to systematically dispatch robots to unexplored clusters. MAPPO, relying on local 24-bin LiDAR ranges and an MLP actor, must learn both mapping and coordinated pathfinding simultaneously from reward gradients.
- **Training budget limits:** At 3.15×10^4 transitions (45 iterations) and 1.05×10^4 transitions (15 iterations), our budget reflects realistic compute constraints for robotic simulation. We cannot rule out that asymptotic training ($> 10^6$ transitions) with curriculum learning or reward normalization would eventually escape the local optimum.

4.2 In-Distribution Performance vs. Zero-Shot Generalization

A critical distinction highlighted by our protocols is in-distribution learning versus cross-world transfer. MAPPO was trained exclusively in `cafe`. Its performance in `warehouse`, `depot`, `office`, and `maze` represents zero-shot transfer across drastically different floor plans (from an open 30×50 m hall to a narrow labyrinth). The Frontier Heuristic maintains consistent advantages across all topologies precisely because its graph-search and Voronoi formulation is topology-agnostic, requiring zero retraining.

5 Conclusion

Under a fully specified, compute-bounded training protocol, MAPPO consistently exhibited low-displacement behavior on multi-robot area coverage across every evaluation episode we ran; we use “consistently observed” rather than “reproducible” deliberately, since we also found that this evaluation harness is not numerically reproducible at the level of individual episode outcomes (Section 3.3.1), even though the qualitative behavior was. Attempting to isolate the failure’s cause via a three-arm reward ablation evaluated at matched statistical power ($n=9$ episodes per arm after discovering and correcting for this non-determinism), we found that strengthening the discovery bonus produced a statistically supported coverage gain (mean ACR 28.0% $\rightarrow$ 40.5%, Welch’s $t = 4.04$) via a mechanism other than increased displacement – revising our original penalty-avoidance-only hypothesis – while weakening the collision penalty produced a smaller, statistically unsupported gain (mean ACR 28.0% $\rightarrow$ $31.0\% \pm 4.5$, $t = 1.48$) that does not clearly exceed the measurement noise of our own evaluation harness, a limitation we discovered only while reconciling an internal inconsistency during this revision, corrected by re-evaluating every arm rather than leaving the discrepancy unresolved, and report in full rather than concealing. Nonetheless, the classical Frontier Heuristic retained superior performance across five diverse environments ($1.2\times$ to $2.0\times$ margin on single-pass estimates now known to carry unquantified uncertainty) without requiring policy training. For roboticists operating under bounded compute budgets, hand-engineered structural coordination remains, on this evidence, a more sample-efficient and more reliably measured

choice than model-free MARL evaluated under a single-episode-per-condition protocol; we recommend that any follow-up work adopt repeated-episode evaluation per condition as a baseline requirement, a methodological lesson this paper learned the hard way.

A Infrastructure Faults Encountered During Training

Two infrastructure faults were encountered and resolved during this work. First, a segmentation fault inside `torch.optim.Adam.__init__` under Ray actor threads was traced to a PyTorch TorchDynamo compilation hook interacting with Ray’s threading model; disabling TorchDynamo (`TORCHDYNAMO_DISABLE=1`) resolved it. Second, training halted at iteration 34/45 when a host Wi-Fi disassociation event caused Ray’s distributed control plane (bound by default to the host’s dynamic IP rather than loopback) to lose connectivity to its own coordination service; binding Ray explicitly to `127.0.0.1` and capping its object store in-process resolved this and allowed training to resume and complete to iteration 45. Neither fault altered the training configuration, reward function, or evaluation protocol reported in the main text, and we verified the resumed run’s hyperparameters and reward coefficients against the pre-crash configuration; we did not, however, independently verify that optimizer or RNG internal state was bit-identical across the interruption, and note this as a residual, unquantified source of uncertainty in the 45-iteration run specifically (it does not affect the Protocol C ablation runs, none of which were interrupted). Both faults are reported for reproducibility.

B Software Versions

Component	Version
OS	Ubuntu 24.04
ROS 2	Jazzy
Gazebo	Harmonic (gz-sim 8)
Python	3.12.3
PyTorch	2.12.0+cu130
Ray / RLlib	2.55.1
Gymnasium	1.2.2
PettingZoo	1.26.1
OSQP	1.1.3
NumPy	2.5.1

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